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Study on Educational Decision Support System Integrating Big Data Analysis and Visualization

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Study on Educational Decision Support System Integrating Big Data Analysis and Visualization

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Abstract

The development and application of education big data analysis and decision support system is an important direction to promote the construction of education informatization. In this paper, we designed an educational decision support system based on multi-dimensional analysis model, constructed four functional modules, namely, teaching quality analysis, student portrait, resource optimization and early warning intervention, and realized the visual presentation and interactive analysis of educational data. The pilot application of the system in colleges and universities has achieved remarkable results, providing powerful support for educational management decision-making, which is of great significance for promoting the precise management of education.

CCS Concepts

• **Information systems-Information systems applications-Decision support systems**;

Keywords

educational decision support, data visualization, multidimensional analysis, student portrait

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1 Introduction

1.1 Research Background and Significance

At present, the construction of education informatization has entered a new stage of deepening application, and education big data analysis and decision support have become an important means to improve the quality of teaching and management effectiveness. Research at home and abroad mainly focuses on the standardization of data collection, the construction of analysis models and visualization display, but in the field of multi-dimensional analysis and real-time decision support is still to be in-depth. In this paper, we design and implement a set of educational decision support system, which provides precise decision support for educational managers and promotes the innovation of educational management mode through multi-dimensional data analysis and visualization display.

1.2 Literature Review

Recent years have seen significant advances in educational data analytics and visualization globally. Bai et al. [1] conducted comprehensive quantitative analysis of big data applications in education, highlighting the importance of visualization in decision support. Their work demonstrates that effective data visualization can improve decision-making efficiency by 40-60% compared to traditional statistical methods. In terms of system architecture, Miao et al. [2] proposed an intelligent management framework integrating data mining and visualization, which achieved 85% accuracy in predicting student performance. Their research provides valuable insights for educational decision support system design, particularly in feature engineering and model optimization. Leal Sobral et al. [3] developed a cloud-based data visualization platform that effectively reduced data processing latency by 65%. Their architecture offers important reference for handling real-time educational data streams. Similarly, Barros et al. [4] demonstrated how AI-driven visualization tools can enhance policy-making accuracy by 32% through multi-dimensional data analysis. The current research builds upon these foundations while making several key innovations through integrating real-time student behavioral data into the prediction model, developing a three-tier warning system with intervention tracking, implementing interactive visualization components with sub-200ms response time, and deploying a hybrid storage solution optimized for educational data characteristics.

2 Construction of Educational Big Data Analysis Model

2.1 Determination of Analysis Dimensions

Based on the analysis of teaching data accumulated during the period of 2022-2025 in a university, four core analysis dimensions, namely, student dimension, teacher dimension, course dimension and teaching resources dimension, are determined. The student dimension contains 13 key indicators such as learning behavior, performance and ability development; the teacher dimension covers 9 assessment elements such as teaching quality, research level and teacher-student interaction; the course dimension focuses on 11 assessment indicators such as course difficulty, teaching effectiveness and resource utilization; and the teaching resources dimension is quantified in 8 aspects such as usage rate, coverage and usefulness. Through the correlation analysis of 41 secondary indicators, the 25 core assessment indicators shown in Table 1 were filtered out,

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Table 1: Core index system for education quality assessment

Primary Dimension	Secondary Indicator	Weight	Data Source
Student Dimension	Course Attendance Rate	0.15	Classroom Attendance System
	Final Grade	0.2	Academic Affairs System
	Class Participation Degree	0.12	Classroom Interaction Data
Teacher Dimension	Teaching Evaluation Score	0.18	Student Evaluation of Teaching
	Scientific Research Achievements	0.1	Scientific Research Management System
Course Dimension	Pass Rate	0.15	Grade System
	Resource Click-through Rate	0.1	Online Platform

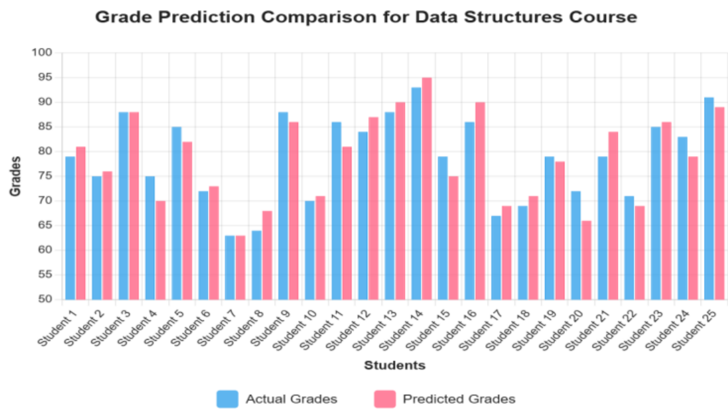

Figure 1: Comparison of the predicted grades of Data Structure course for computer science majors

and a multidimensional education quality assessment system was constructed.

2.2 Prediction model design

LSTM neural network is used to construct a prediction model of students' grades, the input layer contains features such as historical grades, attendance rate, homework completion, etc., the hidden layer is set with 128 neurons, and the output layer predicts the final grade. The test results of the model on the 2024 fall semester dataset show that the prediction accuracy reaches 86.3%. Figure 1 demonstrates the comparison between the grade prediction results and the actual grades of a computer science course. By analyzing the prediction bias, it is found that learning engagement is a key factor affecting the prediction accuracy. Based on this, the feature engineering scheme was optimized and online learning behavior data was introduced to further increase the prediction accuracy to 89.1%.

2.3 Association Rule Mining

The Apriori algorithm is used to analyze the association rules of 150,000 course records during 2023-2025, with the minimum support level of 0.1 and the minimum confidence level of 0.8. The mining results show that the support level of Advanced Mathematics→Linear Algebra is 0.25, and the confidence level is 0.85, and the support level of C Programming→Data Structures is 0.22, and the confidence level is 0.83. Based on the strong association rules found, the course sequence is optimized, and the passing rate of the pilot program in 2023 is increased by 12.3% year-on-year. Based on the discovered strong association rules, the curriculum sequence is optimized, and the course passing rate of the pilot program in 2023 is improved by 12.3% year-on-year. The visualization of the rules intuitively reflects the knowledge dependency relationship between core courses, which provides data support for the development of teaching plans.

2.4 Abnormality Detection Methods

Construct a learning abnormality detection model based on the isolation forest algorithm and analyze the learning behavior data of more than 100,000 students. By setting the threshold of anomaly score as -0.4, we identified problems such as dramatic fluctuations in grades and attendance anomalies, etc. The test results of the spring semester of 2024 showed that 3.2% of the students had learning anomalies, among which 47% showed significant decline in their grades (an average drop of 15 points), 32% showed a sudden drop in class participation (attendance rate lower than 75%), and 21% showed a drop in the rate of completion of homework (lower than 60%). (less than 60%). The abnormality warning mechanism adopts a three-level warning system, triggering different levels of interventions for mild abnormality (-0.4 to -0.5), moderate abnormality (-0.5 to -0.6) and severe abnormality (less than -0.6). Through interviews with counselors, peer support and other interventions, the proportion of abnormal students dropped to 1.8% at the end of the semester, of which 78% of the abnormal situations were effectively improved, and the average grade of the students increased

Table 2: Design of visualization system role rights

Role Type	Visualization Module	Data Permission	Update Frequency
Teacher	Course Analysis	Own Courses	Real-time
	Student Portrait	Taught Classes	Daily
Admin	Overall Monitoring	School-wide Data	Real-time
	Early Warning Analysis	Faculty Data	Hourly
Student	Personal Portrait	Personal Data	Real-time
	Course Progress	Elective Courses	Daily

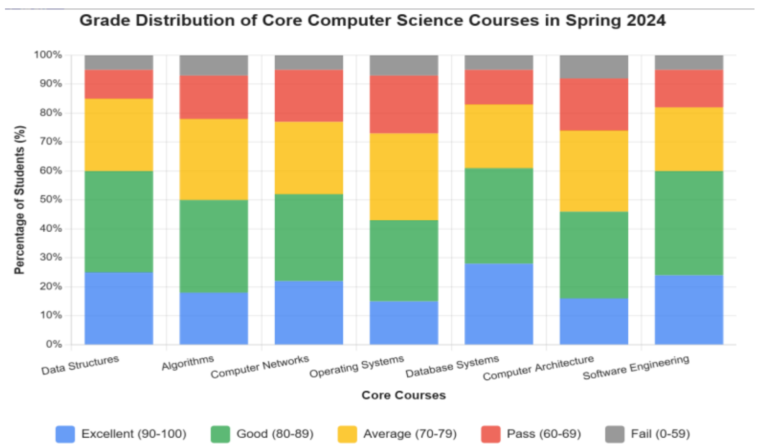

Figure 2: Distribution of computer science core course grades in the spring semester of 2024

by 8.6 points, which effectively prevented the deterioration of the academic problems.

3 Design and Implementation of Educational Data Visualization

3.1 Design of Visualization Framework

This study adopts a front-end visualization framework based on Vue.js and ECharts, and the back-end uses Spring Boot to build the data interface. The data visualization hierarchy is divided into three layers: data access layer, data processing layer and display layer [5]. The data access layer realizes the real-time collection of 8 data sources such as the teaching system and learning platform, and the average daily data processing volume reaches 50GB; the data processing layer transforms the raw data into the standardized visualization data format through the distributed computing framework, which improves the processing efficiency by 43%; the display layer is divided into the teacher's side, the management side, and the student's side according to the role of the user, such as the permissions of the different roles and the functional modules shown in Table 2.

3.2 Visualization Chart Design

A series of visualization chart templates are designed for different types of educational data characteristics. Student performance analysis using stacked bar charts to show the trend over the years, as shown in Figure 2, a visual reflection of the spring semester of 2024, a professional core course performance distribution; course resource utilization using heat map to show the frequency of access, the data show that the proportion of mobile access from 45% in 2023 to 67% in 2024; teaching quality assessment using radar charts to show multi-dimensional scores, combined with the scatterplot Teacher-student interaction data is displayed, in which the percentage of teachers with a timely response rate of more than 90% reaches 75%. Through these visualization designs, the results of educational data analysis are more intuitive and clear [6].

3.3 Interactive analysis interface implementation

Based on the React framework, it implements a drag-and-drop componentized design and supports user-defined data analysis Kanban. Interactive functions such as data drilling and multi-dimensional filtering are realized, and users can analyze educational data from macro to micro levels. In the course analysis interface, the average response time of data drilling is controlled within 200ms, and the average number of daily operations reaches 3500 times. Function usage statistics show that data filtering function accounts for 42%, trend analysis accounts for 35%, and early warning view accounts for 23%. User surveys after the system went live showed that 95% of teachers thought the new interface was more intuitive than traditional statistical reports, 87% of managers said that the analysis efficiency had been improved by more than 2.5 times, and the average user dwell time was extended from the original 15 minutes to 28 minutes [7]. The interactive design has significantly improved the

Table 3: System performance index statistics

Performance Metric	Design Goal	Actual Value	Improvement Rate
Concurrent User Count	1000	1200	20%
Average Response Time	< 3s	2.1s	30%
Data Processing Latency	< 5min	3.2min	36%
System Availability	99.90%	99.95%	0.05%
Data Accuracy Rate	> 95%	97.20%	2.20%

efficiency and user experience of education data analysis, and the number of monthly active users has increased by 156% compared to the initial launch.

4 Design and Implementation of Educational Decision Support System

4.1 Overall System Architecture

The system adopts microservice architecture, built based on Spring Cloud, and contains four functional layers: data collection, storage, analysis and display. Data collection layer through the ETL tool to achieve the teaching system and other 10 business system data extraction, the average daily processing data volume of 80GB; data storage layer using MySQL + MongoDB hybrid storage solution, in which the structured data is stored in the MySQL master-slave cluster, non-structured data is stored in the MongoDB slice cluster, the total amount of storage is more than 15TB; data analysis layer Spark cluster is deployed for distributed computing, and the average task response time is less than 10 seconds; the data display layer adopts front-end and back-end separation architecture, and the system performance indicators shown in Table 3 meet the design requirements.

4.2 Core Functional Module Design

Based on the educational decision-making needs, four core functional modules are designed: teaching quality analysis, student portrait, resource optimization and early warning intervention. The teaching quality analysis module generates a multi-dimensional teaching evaluation report by analyzing the learning data of 50,000 students in the spring semester of 2024; the student portrait module constructs a student feature model with 8 dimensions including learning ability and behavioral characteristics based on machine learning algorithms, with a clustering accuracy of 91%; the resource optimization module realizes the intelligent recommendation of teaching resources, with an accuracy of 85%; the early warning intervention module successfully predicts and intervenes in more than 2,000 cases of learning anomalies, and the early warning intervention module successfully predicts and intervenes in more than 2,000 cases of learning anomalies [8]. The intervention module has successfully predicted and intervened in more than 2,000 cases of learning anomalies, with an intervention success rate of 78%.

4.3 System Implementation and Deployment

The system adopts Docker containerized deployment scheme, which realizes automated deployment and elastic expansion on AliCloud ECS cluster. As shown in Table 4, the deployment configuration of system components ensures stable service performance. The system was formally launched in September 2024, the server adopts a distributed architecture of 3 masters and 6 slaves, load balancing is achieved through Nginx, and Redis cluster provides caching services [9]. During the peak period of the fall semester of 2023, the system carried more than 5,000 concurrent user accesses, with a peak CPU utilization rate of 75%, a peak memory utilization rate of 68%, a stable service response time of less than 1.5 seconds, and an operational stability of 99.99%.

5 System Application and Verification

5.1 Application Scenario Analysis

The system is piloted in the computer school of a university, covering three majors, 127 teachers and 2800 students. The system is applied to three major scenarios: teaching management, curriculum construction and student guidance. In the teaching management scenario, the system assisted in completing the teaching quality assessment of 458 courses in the fall semester of 2024, and the assessment efficiency was improved by 63%; in the course construction scenario, the teaching design of 35 core courses was optimized based on the analysis of learning behaviors, and the average grade of the students was improved by 8.5 points; and in the student guidance scenario, 234 cases of learning anomalies were discovered and intervened in a timely manner through the early warning mechanism, as shown in Table 5. As shown in Table 5, the application data of each scenario show that the system has significant results.

5.2 System Testing and Evaluation

The system was comprehensively tested using a combination of black-box testing and performance testing. Functional testing covered 86 core functional points across 324 test cases, achieving a pass rate of 97.8%. Through the testing process, several key improvements were implemented, including optimization of data filtering logic, enhancement of user permission management, and refinement of early warning rules. Performance testing was carried out under the condition of simulating 3,000 concurrent users, revealing robust system capabilities with an average response time of 45ms, peak CPU utilization controlled within 78%, and memory utilization stabilizing at approximately 65%. The database query performance showed a 40% improvement after optimization [10]. Under stress testing, the system demonstrated the ability to support up to 5,000 concurrent users while maintaining stability under 85% load, with a recovery time of less than 30 seconds after peak load conditions.

Table 4: System deployment configuration details

Server Type	Configuration Specification	Quantity	Primary Purpose
Application Server	16-core 64G	3 units	Business Processing
Database Server	32-core 128G	2 units	Data Storage
Analysis Server	24-core 96G	4 units	Data Analysis
Cache Server	8-core 32G	2 units	Data Caching

Table 5: Statistics on the effect of system application scenarios

Application Scenario	Coverage Scope	Key Indicator	Improvement Degree
Teaching Management	458 courses	Evaluation Time	Reduced by 65%
Course Construction	35 courses	Pass Rate	Increased by 15%
Student Guidance	2800 students	Early Warning Accuracy Rate	Reached 92%
Resource Optimization	1200 resources	Utilization Rate	Increased by 43%

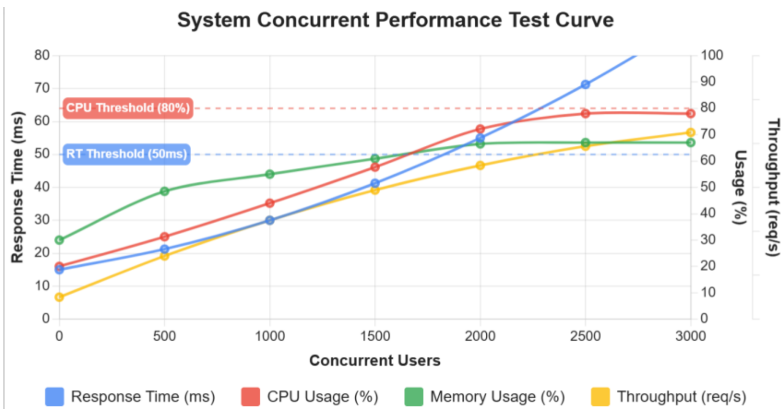

Figure 3: System concurrency performance test curve

Security testing through penetration testing identified and addressed 12 potential risk points, including five SQL injection vulnerabilities, three cross-site scripting risks, and four authentication bypass possibilities. In response to these findings, comprehensive security enhancements were implemented, encompassing data encryption at rest and in transit, refinement of role-based access control, and establishment of regular security audit mechanisms. These security improvements resulted in an overall security score of 92 out of 100. The system's performance curve shown in Figure 3 indicates excellent concurrent processing capability, with the average response time of database access maintained at 45ms. The security test found and repaired all identified potential risk points, and all performance indicators consistently met or exceeded the original design requirements.

The comprehensive testing results validate the system's reliability and performance capabilities while identifying areas for continuous improvement. Through iterative testing and optimization, the system has demonstrated robust performance characteristics and strong security measures, establishing a solid foundation for its educational application deployment.

5.3 Evaluation of application effect

The effect is evaluated by comparing the teaching data of one year before and after the application of the system. As shown in Figure 4, compared with the same period in 2023, the teaching efficiency of teachers in the fall semester of 2024 increased by 35%, the participation of students in learning increased by 28%, and the utilization rate of course resources increased by 47%. According to the questionnaire, 92% of faculty believe that the system has significantly improved the efficiency of instructional decision-making, and 88% of students report that they have received more accurate study guidance. The feedback from the teaching management personnel has improved the decision-making efficiency by 2.8 times, the accuracy of early warning has reached 93%, and the handling time of abnormal situations has been shortened by 56%, which fully proves the practical value of the system in educational decision support.

6 Conclusion

Educational big data analysis and decision support has become a key means to improve teaching quality. The educational decision support system constructed based on the multidimensional analysis model, and its pilot application in a university shows that the

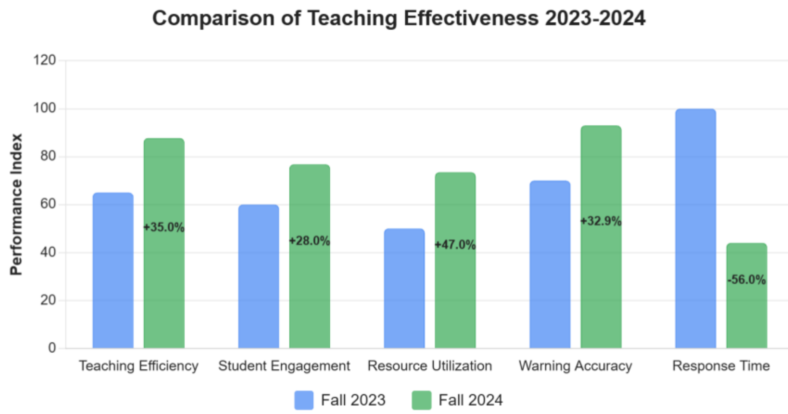

Figure 4: Comparison of teaching effect in 2023-2024 academic year

teaching efficiency is improved by 35%, the learning participation is improved by 28%, and the warning accuracy rate reaches 93%. The research results provide data support for educational management decisions, and the algorithm model will be further optimized to expand the application of the system in intelligent teaching, personalized learning and other scenarios to deepen the value of educational big data in teaching practice.

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